

SANTOSH THAKUR

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer with 6+ years of experience building enterprise-scale applications with **React.js, Next.js, TypeScript, Python, and Generative AI**. Experienced in building production AI applications using **OpenAI, Azure OpenAI, AWS Bedrock, RAG, LangChain, multi-agent systems, streaming responses, and vector search**. Strong frontend expertise in scalable React architecture, Micro Frontends, performance optimization, accessibility, and high-traffic applications, combined with hands-on backend development using **FastAPI/Flask, REST APIs, PostgreSQL, and cloud-native services**.

CORE SKILLS

Front-End: React, Next.js, JavaScript (ES6+), TypeScript, Context API, Redux Toolkit, Angular, NgRx, GraphQL, Apollo Client, Micro Frontends, Storybook, SCSS, Tailwind CSS, Material UI, Core Web Vitals, Accessibility

Backend/Integration: Python, FastAPI, Flask, Node.js, REST APIs, GraphQL, WebSockets, Server-Sent Events (SSE), JWT

AI & Modern Web:: OpenAI APIs, Azure OpenAI, AWS Bedrock, RAG, LangChain, Multi-Agent Systems, LLM Integration, Prompt Engineering, Streaming Responses, Structured Outputs, Function Calling, Embeddings, Vector Search, Azure AI Search

AI & Modern Web: OpenAI Responses API, OpenAI SDK Integration, LLM Integration, AI Chat Interfaces, Prompt Engineering, Streaming Responses, Structured Outputs, Function Calling, AI Workflow Integration, Generative AI Applications, GitHub Copilot, Cursor AI, Claude AI

Cloud & DevOps: AWS, Azure, AWS Lambda, S3, DynamoDB, AppSync, AKS, Docker, Kubernetes, KEDA, CI/CD, CloudFormation

Databases & Testing: PostgreSQL, MongoDB, Jest, React Testing Library, Playwright, Selenium, SonarQube

PROFESSIONAL EXPERIENCE

Senior Associate Consultant – Software Engineer / Infosys Ltd., India

November 2025 – Present

- Develop enterprise applications using **React.js, TypeScript, Python, and FastAPI**.
- Build reusable UI components and integrate frontend applications with REST APIs and backend services.
- Contribute to AI-enabled application workflows and integrate modern AI development tools into engineering workflows.
- Improve application performance, accessibility, and overall reliability.
- Implement automated testing using Playwright and Selenium.
- Work with **RAG, document processing, vector search, and AI-assisted application patterns** for enterprise use cases.
- Use GitHub Copilot and AI-assisted engineering tools for implementation, debugging, testing, and documentation.
- Participate in code reviews, Agile development, and CI/CD deployments.

Technologies Used: React.js, Next.js, TypeScript, Python, FastAPI, REST APIs, OpenAI, RAG, AI Workflows, Redux Toolkit, Playwright, Selenium, GitHub Copilot

Software Developer – Full-Stack Engineer / BagConvergence

March 2021 – November 2025

- Built and maintained enterprise-scale **React.js and Next.js applications serving 5M+ daily users** across multiple products.
- Designed full-stack application architecture spanning **frontend systems, API integration, real-time communication, and cloud services**.
- Built reusable component libraries and **Micro Frontends** supporting independently developed and deployed business modules.
- Integrated **GraphQL and REST APIs** and optimized data fetching, caching, and state management for high-volume applications.
- Built WebSocket-based real-time dashboards supporting **200K+ concurrent users**.
- Improved application performance by **50%+** through code splitting, lazy loading, memoization, caching, and image optimization.
- Contributed to **Python backend integrations, Docker-based services, AWS deployments, and CI/CD pipelines**.
- Achieved **90%+ mobile performance scores** through Core Web Vitals and application optimization.

Technologies Used: React.js, Next.js, TypeScript, Javascript, Redux Toolkit, Angular, GraphQL, Python , REST APIs, WebSockets, Docker, AWS, CI/CD

Front-End Developer / ModernVastu Research & Remedy Pvt. Ltd.

April 2019 – March 2020

- Developed responsive React web applications with Tailwind CSS and Material UI and integrated RESTful APIs for seamless data flow

Technologies Used: React, JavaScript, jQuery, HTML5, CSS3, Bootstrap, Tailwind CSS, Material-UI, RESTful APIs, Github

KEY PROJECT HIGHLIGHTS

AI Chat Platform: Built AI-powered chat interfaces using OpenAI APIs. Implemented streaming responses. Developed reusable React components. Used Prompt Engineering to improve AI response quality. Integrated secure REST APIs.

Internal Contracts AI: Built a production **RAG-based contracts Q&A platform** from zero to production in five weeks using FastAPI, PostgreSQL, Qdrant, LangChain, and AWS Bedrock. Implemented JWT security, **SSE streaming, multi-page citations, asynchronous indexing, deduplication, and intent-based routing**.

Headless CMS Entertainment Platform: Built with Next.js + WordPress VIP. Automated content workflows, optimized image delivery, and implemented advanced SEO.

EDUCATION

Master of Computer Applications (MCA) / Maharaja Agrasen Himalayan Garhwal University (2020 – 2022)

Bachelor of Computer Applications (BCA) / IEC University (2015 – 2018)

